

Python programming

How to run this unit, lesson by lesson

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Start here

13 lessons. A realistic estimate is **39 teaching hours**: most lessons are one hour, and a lesson with thirty or more activities takes two or three. The estimate counts transitions and assumes you teach from the front for the first fifteen minutes of each lesson.

Teach them in the order below. The order is the retrieval schedule as well as the teaching one: each starter asks about the lesson before it.

#	LESSON	HOURS	PLAN
1	Printing and variables pr-101	3	Lesson_Plans/PR_L01_LessonPlan.pdf
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Preparation and what you need

1. **Class logins, before the first lesson.** On the teacher dashboard (`teacher.html`), open **Class logins**, enter the class and the usernames one per line, and print the login cards. A pupil cannot sign in without one.
2. **Devices.** A laptop, a tablet or a phone will do. A headset is optional everywhere. The browser needs to be current; the experiences use WebGL.
3. **Print the worksheets** for each lesson. They are in this pack as `.docx` to edit and `.pdf` to print.
4. **Open each experience once yourself** on the teaching machine, so the images are cached before thirty pupils ask for them at the same moment.

Where progress is kept. In this build `APP_CONFIG.backendUrl` is empty, so logins, login cards and progress are stored on the device a pupil used, not on a server. The teacher dashboard says so itself. Keep a class on the same

devices within a unit, and treat the paper worksheet as the record that travels with the pupil.

Pupils sign in at `topics.html` with the username and PIN on their card, choose the topic and then the lesson.

Using the three resources together

Slides for the front of the room; **the experience** for reading, deciding and being marked; **the worksheet** for writing and for the record. The rule that holds them together is write first, then answer at the badge.

Do not teach the station walls and then send pupils to read the same walls. Teach the first two or three from the slides and let them read the rest; the walls are there so a pupil can re-read what you said, not so you can say it twice.

Evidence, checking and common difficulties

Collect three things from every lesson: the starter (written), the key facts and station challenges on the worksheet (written), and the score screen breakdown (copied onto the worksheet by the pupil as soon as it appears). The score screen shows the marks per station, which is what tells you where to reteach; a single total does not.

The teacher dashboard shows progress by pupil and the class's weakest stations, averaged on first attempts. Use the weakest-station list to choose what to reteach rather than asking who found it hard.

The lesson plans, in teaching order

Each lesson has its own plan as a separate PDF in Lesson_Plans/ of this pack, so you can print the one you are teaching. What follows is the summary of each; the plan carries the timings, the hinge question, the station-by-station delivery and the adaptations.

1. Printing and variables pr-l01

Why must a program decide the data type of a value before it can do anything useful with it?

OUTCOMES	know how to print lines of output in the order a task asks for; know how to hold a value in a variable or a constant, and how to read one with <code>input()</code> ; understand why a value that has been typed in must be cast before it can be calculated with
STATIONS AND ACTIVITIES	6 stations, 56 activities, 108 marks (16 guided, 40 independent, 6 challenge)
RESOURCES	PR_L01_PrintingAndVariables_Lesson.pptx, PR_L01_PrintingAndVariables_Worksheet.docx
PLAN	Lesson_Plans/PR_L01_LessonPlan.pdf

2. Arithmetic pr-l02

How do you choose between `/`, `//` and `%` when the answer to a calculation has to come out as a whole number?

OUTCOMES	know which arithmetic operator each kind of calculation needs; understand the difference between <code>/</code> , <code>//</code> and <code>%</code> and what each one hands back; know how to round a result, and how to draw a random number from a seed
STATIONS AND ACTIVITIES	6 stations, 54 activities, 104 marks (16 guided, 38 independent, 6 challenge)
RESOURCES	PR_L02_Arithmetic_Lesson.pptx, PR_L02_Arithmetic_Worksheet.docx
PLAN	Lesson_Plans/PR_L02_LessonPlan.pdf

3. Selection pr-l03

Why does the order the branches are written in change what an `if ... elif ... else` actually does?

OUTCOMES	know how to write a condition using the comparison operators; understand how <code>if</code> , <code>elif</code> and <code>else</code> choose between several outcomes; know how to join two conditions together with <code>and</code> , <code>or</code> and <code>not</code>
STATIONS AND ACTIVITIES	6 stations, 55 activities, 110 marks (14 guided, 41 independent, 6 challenge)
RESOURCES	PR_L03_Selection_Lesson.pptx, PR_L03_Selection_Worksheet.docx

PLAN	Lesson_Plans/PR_L03_LessonPlan.pdf
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4. Count-controlled loops pr-104

How do you decide what to put inside range() so that a loop counts exactly the values you want?

OUTCOMES	know how to write a for loop that repeats a block a set number of times; understand how range() decides which values the loop counter takes; know how to build a running total inside a loop, and how to put one loop inside another
STATIONS AND ACTIVITIES	6 stations, 55 activities, 109 marks (17 guided, 38 independent, 6 challenge)
RESOURCES	PR_L04_CountControlledLoops_Lesson.pptx, PR_L04_CountControlledLoops_Worksheet.docx
PLAN	Lesson_Plans/PR_L04_LessonPlan.pdf

5. Condition-controlled loops pr-105

How do you decide between a for loop and a while loop before you write a single line of code?

OUTCOMES	know how to write a while loop and what makes it stop repeating; understand how a validation loop keeps asking until the value typed in is acceptable; know how a sentinel value ends a loop when the amount of data is not known in advance
STATIONS AND ACTIVITIES	6 stations, 55 activities, 114 marks (13 guided, 42 independent, 6 challenge)
RESOURCES	PR_L05_ConditionControlledLoops_Lesson.pptx, PR_L05_ConditionControlledLoops_Worksheet.docx
PLAN	Lesson_Plans/PR_L05_LessonPlan.pdf

6. String handling pr-106

Why are the characters of a string numbered from 0 rather than from 1?

OUTCOMES	know how to find the length of a string and the character at a given position; understand how a slice takes part of a string, and why its end position is left out; know how to change case, search inside text, and move between a character and its code
STATIONS AND ACTIVITIES	6 stations, 44 activities, 84 marks (15 guided, 29 independent, 6 challenge)
RESOURCES	PR_L06_StringHandling_Lesson.pptx, PR_L06_StringHandling_Worksheet.docx
PLAN	Lesson_Plans/PR_L06_LessonPlan.pdf

7. Lists pr-107

Why is the last item of a list always at index len(list) - 1?

OUTCOMES	know how to create a list and read one item out of it by its index; understand how to traverse a list in order to total, count or compare its items; know how to add an item onto a list and how to take one out again
STATIONS AND ACTIVITIES	6 stations, 44 activities, 93 marks (12 guided, 32 independent, 6 challenge)
RESOURCES	PR_L07_Lists_Lesson.pptx, PR_L07_Lists_Worksheet.docx
PLAN	Lesson_Plans/PR_L07_LessonPlan.pdf

8. Two-dimensional lists pr-108

Which index comes first when a program reads a cell, and why does swapping the two over give a different answer?

OUTCOMES	know how to build a two-dimensional list as a list of lists and fill it from input; understand how a row index and a column index together pick out one cell; know how to write nested loops that print a grid and total any row or column of it
STATIONS AND ACTIVITIES	6 stations, 43 activities, 89 marks (12 guided, 31 independent, 6 challenge)

RESOURCES	PR_L08_TwoDimensionalLists_Lesson.pptx, PR_L08_TwoDimensionalLists_Worksheet.docx
PLAN	Lesson_Plans/PR_L08_LessonPlan.pdf

9. Subprograms pr-109

When should a subprogram return a value rather than print one?

OUTCOMES	know how to define a subprogram with def and how to call it; understand how a parameter carries a value into a subprogram and how return carries one back out; know how to write a subprogram that uses a variable of its own, and one that changes a shared variable
STATIONS AND ACTIVITIES	6 stations, 43 activities, 87 marks (12 guided, 31 independent, 6 challenge)
RESOURCES	PR_L09_Subprograms_Lesson.pptx, PR_L09_Subprograms_Worksheet.docx
PLAN	Lesson_Plans/PR_L09_LessonPlan.pdf

10. File handling pr-110

Why does the mode a file is opened in matter more than anything else on that line?

OUTCOMES	know how to open a text file, read what is in it and close it again; understand what write mode does to a file, and how append mode differs from it; know how to split one line of a file into its separate fields and use them
STATIONS AND ACTIVITIES	6 stations, 43 activities, 87 marks (12 guided, 31 independent, 6 challenge)
RESOURCES	PR_L10_FileHandling_Lesson.pptx, PR_L10_FileHandling_Worksheet.docx
PLAN	Lesson_Plans/PR_L10_LessonPlan.pdf

11. Searching and sorting pr-111

Why is a binary search so much faster than a linear search on a long list, and what does using one cost you?

OUTCOMES	know how to code a linear search and a binary search, and report the position each one finds; understand how a bubble sort, an insertion sort and a merge sort each put a list in order; know how to count the comparisons an algorithm makes, in order to compare two of them
STATIONS AND ACTIVITIES	6 stations, 45 activities, 88 marks (19 guided, 26 independent, 6 challenge)
RESOURCES	PR_L11_SearchingAndSorting_Lesson.pptx, PR_L11_SearchingAndSorting_Worksheet.docx
PLAN	Lesson_Plans/PR_L11_LessonPlan.pdf

12. Validation and authentication pr-112

Why is refusing bad data better than letting a program stop with an error?

OUTCOMES	know how to code a presence check, a type check and a range check; understand how a length check and a format check differ, and why the length is tested first; know how to write a validation loop, and an authentication check that allows a set number of attempts
STATIONS AND ACTIVITIES	6 stations, 46 activities, 90 marks (18 guided, 28 independent, 6 challenge)
RESOURCES	PR_L12_ValidationAndAuthentication_Lesson.pptx, PR_L12_ValidationAndAuthentication_Worksheet.docx
PLAN	Lesson_Plans/PR_L12_LessonPlan.pdf

13. SQL queries pr-113

Which clause of a query decides which fields come back, and which clause decides which records?

OUTCOMES	know how SELECT and FROM choose the fields and the table a query reads; understand how WHERE decides which records a query hands back; know how to join conditions with AND and OR, and how to put results in order with ORDER BY
STATIONS AND ACTIVITIES	6 stations, 44 activities, 79 marks (15 guided, 29 independent, 6 challenge)
RESOURCES	PR_L13_SQLQueries_Lesson.pptx, PR_L13_SQLQueries_Worksheet.docx
PLAN	Lesson_Plans/PR_L13_LessonPlan.pdf

Teacher answers

Model answers and marking guidance are not in this guide and not on the lesson plans. A plan is printed and left on a desk; an answer sheet is not. Where verified answers exist for this unit they are in the Teacher_Answers/ folder of this pack. Where they do not, the pack says so rather than shipping an empty file.

For the programming lessons there is no single model answer to print: a program is marked by running it against its test cases, and many different correct programs pass. The tests themselves are the marking guidance, and they are in the teacher answers.

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